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TEC–602

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

DIGITAL SIGNAL PROCESSING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the linearity and time-shifting properties of the Discrete-Time Fourier Transform (DTFT), elucidating how each property manifests in the mathematical representation of signals in the frequency domain. (CO1)

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OR

(b) Compute the DFT of the 3-point sequence : (CO1)

$$x(n) = \{2, 1, 2\}$$

using direct method.

2. (a) Determine the circular convolution of following sequence : (CO1)

$$y(m) = x(n) \odot h(n).x(n) = \{1, 2, 3, 1\}$$

and

$$h(n) = \{4, 3, 2, 2\}$$

OR

(b) Describe the connection between the Discrete Fourier Transform (DFT) and the Z-transform, highlighting the relationship between these two mathematical transformations. (CO1)

3. (a) Compute the length-4 sequence from its DFT which is given by : (CO1)

$$X(k) = \{4, 1 - j, -2, 1 + j\}$$

OR

(b) Determine the DTFT of the following sequence : (CO1)

(i) $x(n) = u(n)$

(ii) $x(n) = a^n u(n)$

4. (a) Describe the butterfly diagram for an 8-point DIT FFT, including the initial sequence order, bit-reversal permutation, and computations in 2-point, 4-point, and final 8-point stages. Emphasize the flow of data and transformations at each step. (CO2)

OR

(b) Explain how the FFT algorithm contributes to the efficiency of computing the DFT. Determine the DIFFFT of the following sequence. (CO2)

$$x(n) = \{0,1,0,1,0,1,0,1\}$$

5. (a) Determine the DFT of the four-point sequence : (CO2)

$$x(n) = \{0, 1, 2, 3\}$$

using matrix method.

OR

(b) Using FFT and IFFT, determine the output of the system if input $x(n)$ and impulse response $h(h)$ are given as under : (CO2)

$$x(n) = \{2, 2, 4\}$$

and $$h(n) = \{1, 1\}.$$